

The archive of Haghia Triada: what was counted, how the account was organised, and what is not recorded

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Summary

Haghia Triada preserves the largest administrative archive of Linear A: 1,108 of the 1,720 documents of the corpus. This paper describes it as an archive and not as a language, under a single procedure: every claim about what a unit does is fixed by distribution against a permutation null with a measured error rate, and filed with the evidence for, the evidence against, and a refutation condition. Six results. **First**, the anatomy of the Minoan receipt compared with six Bronze Age administrations: the same five functions and two absences, with no transmission required. **Second**, the functional inventory, which rises from 65 units fixed one by one to 151 once a class-level function is admitted: the first syllabic unit of a tablet takes a figure 29% of the time against 72% for all others ($p = 8 \cdot 10^{-34}$), so the opening position is the label of the record and not an entry in the account. **Third**, five tablet formats identified by motif discovery against a Markov null, among them the transaction template, which the procedure recovers unprompted. **Fourth**, the list entries, the most numerous class and until now the one without an instrument, fall into four stable groups that recur together across documents (35 pairs against 8.2 under the null), one of which Uchitel and Finkelberg set out by inspection in 1995 while the other three are a different partition of the same material: they are sets of membership and not sequences, and one of them receives grain in multiples of ten with the most regular quantities in the corpus. **Fifth**, allocation is modular: eight of 119 tablets have every figure a multiple of the smallest (null 1.9) and fifty-one repeat one figure three or more times (null 36.0), which describes allocation by quota rather than an inventory of stock. **Sixth**, the inventory declares its scope: 51 of the 65 readings fixed one by one are fixed on the accounting archive, 7 on the votive inscriptions and 4 on the sealings, and a claim fixed on one genre does not transfer to another. **Seventh**, the two classes of sign are bound by a rule: the syllabogram ligatured to a commodity logogram belongs to that commodity in 16 of 26 cases against 25% expected by chance ($p < 0.0005$), which is the acrophonic device Linear B uses, and the ten that circulate are precisely the entries this inventory had filed as general qualifiers. **Eighth**, each commodity has its own metrology: grain is counted in tens (34% multiples of ten against a base rate of 17%) while oil, olives and cyperus fall below that rate, which accounts for the archive of Khania, dominated by cyperus, never summing (0 of 105 tablets against 41 of 205) and recording in fractions. Three facts about the corpus accompany all of this: 276 of 783 syllabic units are broken in every attestation; the archive records no metal and no weaponry, with 1 of 1,480 logogram occurrences against 14.3% in Linear B; and the ceiling of what the internal method fixes with this corpus is functions, not words.

Keywords: Linear A, Haghia Triada, Neopalatial administration, functional inventory, metrology, null models, class-level functions.

Introduction: an archive that is understood and a language that is not read

What is known of Linear A with certainty fits in a sentence: it is the script of a Cretan administration of the sixteenth century BC that counted grain, oil, wine, figs, livestock and people, and added them up. What is not known is what any of it was called. Between those two things lies a space the field has filled with proposed readings and that this paper fills with something else: a measured description of the archive, unit by unit, with what each one does and what would refute it.

The paper has six parts. The first (§1) asks what shape a Minoan receipt has and how it resembles and differs from those of Susa, Uruk, Ebla, Ugarit, Pylos and Knossos. The second (§2) builds the inventory of units with a fixed function and declares the three facts about the corpus that the editions do not state. Two further parts (§2 bis, §2 ter) treat the formats of the tablet and the internal structure of the list entries. The fifth (§3) descends to one file, the aromatics, where the procedure produces class readings and an identification under dispute. The sixth (§4) compares the archive of Khania with that of Haghia Triada and shows they are not the same archive at different sizes. The discussion (§5) states what is fixed, what remains open, and why the number of functions stops where it does.

Part 1. The anatomy of the receipt

1.1 The question

That Minoan writing arises for accounting is not disputed; whether its form is native or learned is. Crete traded with the Levant and with Egypt from the Protopalatial period, its products appear at Ugarit and at Amarna, and from the East come goods whose names Greek later inherited through Minoan (sesame, cumin). The transmission of goods and of words is documented. Was the form of the document transmitted too?

The question can be answered without reading Minoan, because the form of an archive is measurable: which administrative functions it puts into words, and how often. This work measures it in six archives and answers no.

1.2 Method and data

For each corpus the closed class is extracted: the lexical units that recur across many documents and are neither proper names nor commodities. In the lemmatised corpora the class is read from the glossary with its instance counts; in the unlemmatised ones the forms occurring in a minimum number of documents, and across several sites where there is more than one. Each term is then assigned to one of five administrative functions: total and account; transfer (give, receive, dispatch, by the hand of); responsibility and seal; date; measure. The assignment is the field's own in the read corpora, and ours by distribution in Linear A, where three of the terms are additionally fixed by arithmetic.

Corpora: Linear A (GORILA [1] via the LinearA Explorer, 1,098 word occurrences on tablets, 782 types); TCMA, Archival Texts of the Middle Assyrian Period (ORACC, 2,276 tablets,

89,677 lemmatised instances, 2,524 lemmas); The Amarna Letters (ORACC, 305 texts, 29,961 instances); SAA 7 (ORACC, 219 texts, 8,574 instances); BDTNS for Ur III (20,000 tablets sampled from 101,389, 432,915 tokens); CDLI for proto-Elamite (1,594 tablets) and Uruk IV-III (6,387).

1.3 The anatomy is common

All six archives write the entry as unit plus quantity, open with a header, and close with a total in some documents. The proportion of entries with a quantity runs from 0.60 to 0.80. The partition of the lexicon is the same in all: a short class of frequent repeated terms, which are the commodities and the functions, and a long class of unique units, which are the names and places. At Susa, one-sign strings are singletons in 3% of cases and strings of three or more in 87-99%; at Haghia Triada 76% of the vocabulary is singleton; in Ur III and Middle Assyrian the proportion is of the same order once proper names, which both corpora mark explicitly, are discounted.

That anatomy occurs in proto-Elamite and at Uruk, fifteen hundred years and two thousand kilometres from Crete, with no relation between Susa and Crete. It is the shape a record of deliveries takes when it is written down, and it requires no transmission.

The identification of the Minoan terms is not a contribution of this work. Schoep [2] (2002) established them with their function: TE, separated by dividers on both sides and always at the head of a list with one or more logograms; KU-RO, “total”, followed as in Linear B by the sum of the preceding entries; PO-TO-KU-RO, “grand total”; KI-RO, only at Haghia Triada, as deficit, on the argument that when it follows a KU-RO it is always smaller; and SA-RA₂, also exclusive to Haghia Triada, whose mutual exclusion with TE within mixed-commodity tablets suggests the two represent opposite transactions. What is contributed here is not the inventory of terms but the comparison of functional repertoires across six archives.

1.4 The five functions, and the two absences

function	Middle Assyrian (14th-11th c.)	Amarna (14th c.)	Ur III (c. 2000)	Linear A (c. 1450)
total and account	<i>napharu</i> “total” (811), <i>nikkassu</i> “account” (93)	<i>gabbu</i> “totality” (184)	šu-nigin ₂ , nig ₂ -ka ₉	KU-RO (37), PO-TO-KU-RO, DA-I
deficit	(marked by subtraction)	—	la ₂	KI-RO (16)
transfer	<i>nadānu</i> “give” (613), <i>nadnu</i> “given” (401), <i>leqû</i> “take” (365)	<i>nadānu</i> (238), <i>leqû</i> (213)	ba-ti “received”, ba-zi “dispatched”, giri ₃ “by the hand of”	none
responsibility	<i>kunukku</i> “seal”	<i>ṭupšarru</i> (4)	kišib, maškim,	the physical seal;

function	Middle Assyrian (14th-11th c.)	Amarna (14th c.)	Ur III (c. 2000)	Linear A (c. 1450)
and seal	(789), <i>ṭupšarru</i> “scribe” (391), <i>waklu</i> “overseer” (88)		ugula	TE on the nodules; A-DU and KA-PA as headers
date	<i>ūmu</i> “day” (2,599), <i>līmu</i> “year eponym” (1,849), <i>warhu</i> “month” (251)	<i>ūmu</i> (50), <i>šattu</i> (33)	mu, iti, u ₄	none
measure	<i>sūtu</i> (4,459), <i>imēru</i> (4,046), <i>qû</i> (1,779)	mina, shekel (residual)	gur, sila ₃ , gin ₂	its own system of fractions

Two observations. The first concerns coverage: the twenty most frequent units cover 47.6% of the Middle Assyrian corpus, 52.2% of Amarna, 28.3% of Ur III and 13.7% of Linear A. The low Cretan figure does not indicate a poorer functional vocabulary but that commodities there are written with logograms and not with words. Comparing like with like, the closed class is of the same order in all six: between fifteen and twenty-five terms.

The second is what matters. In the nearest contemporary archive, transfer and date are not marginal functions: “give” and “take” together exceed a thousand instances, the year eponym occurs 1,849 times and the day 2,599, which is to say almost every tablet is dated. In Linear A there is no word that behaves as a verb of transfer or as a date. And the absence is not one of identification: of the twenty words that make up the closed class of Linear A, all have a function assigned by position or are frequent names; none is left without a destination that could be a “received” or a “year”.

1.5 What Crete does instead

It delegates both functions to the object and to the cycle.

The seal does the work of the verb. Haghia Triada produces hundreds of nodules and roundels against a modest number of tablets; the roundel carries a sign group, sometimes a commodity logogram, and as many seal impressions as units. It is a complete receipt with no word of transfer: who delivers is in the sign group, what is delivered in the logogram, how much in the impressions, and before whom in the seal, which identifies the receiving official and not the party. Middle Assyrian, which also seals, writes the word *kunukku* 789 times besides; Crete seals and does not write it.

The cycle does the work of the date. An archive without dates cannot keep multi-year accounts or justify deliveries to third parties: it records the current cycle. That agrees with what

archaeology says of Minoan archives, which were remade and recycled and not kept from one year to the next, and with Haghia Triada being the photograph of a single year, the year of the destruction, written by twenty-one scribes working at once.

The same asymmetry appears in a different genre and confirms it. The seven perfume recipes of the Assur archive write the procedure in full: light the fire, cover the bowl, let it stand overnight, sieve at dawn, discard the residue, and repeat the pouring counting the times. The Minoan tablets of the aromatics file at Haghia Triada record the same ingredients with the same structure of quantities and write not one step of the procedure. Assyria writes what is done; Crete, what comes in. The parallel extends to the genre closest to it: PY Un 267 records the delivery of ingredients to the perfumer and writes the verb of delivery, the name of the deliverer, the name and profession of the recipient and the purpose, while the corresponding Minoan tablets write none of those four things.

1.6 Discussion

What the comparison allows one to state. The anatomy of the receipt is universal in the trivial sense: it is the shape any record of deliveries takes, and it appears in administrations without contact. What distinguishes traditions is which functions they verbalise, and there Crete does not resemble the Eastern model precisely in the two functions that model puts in the foreground. Had Minoan accounting been learned from the Levant or Mesopotamia, we should expect to find transfer and date written, because they are the most frequent words of the model; their systematic absence points to native development on a common anatomy.

What it does not allow. It does not say that Crete was ignorant of Eastern practice: the Minoans traded with people who dated and wrote “received”, and could have known the practice and not adopted it, which is a decision and not an ignorance. Nor does it say anything about the language: the form of the document and the language it is written in are independent. And it rests on a small Minoan corpus; the claim that holds is about what the archive of Haghia Triada writes and does not write, and it would weaken if a Minoan archive with dates appeared.

What it suggests for the history of trade. The words that did travel to Crete from the East are commodity words (sesame, with its Semitic reduplication preserved; cumin), and those that travelled from Crete to Greece are commodity words too (cyperus, the sycomore fig, perhaps henna). The products and their names travelled, with the ships; the form did not. It is an informative asymmetry: Bronze Age trade transmitted goods and lexicon easily and administrative practice with difficulty, which fits administration being the domain where each palace protected its own way of doing things.

Two Syro-Levantine archives fall outside for want of machine-readable data, and this should be said precisely because they are the ones nearest to Crete. Alalakh has 771 objects catalogued in CDLI, 523 administrative and 294 from level IV, contemporary with Haghia Triada, but only three with available transliteration, and all three are lexical texts. A pending and decisive check is the archive of Ugarit, Crete’s direct trading partner in the fourteenth century. With Ugarit

measured, the conclusion would move from “the nearest contemporary archive” to “the archive of the trading partner”, which is the strong test. This limitation is of the material and not of the method: the archives of Ugarit and Alalakh are treated here from published descriptions and not from the texts, since no digitised corpora of those archives exist in a form comparable to the others (J. Lauinger, personal communication, 9 September 2026).

Part 2. The functional inventory

2.1 The corpus and what the editions do not say about it

The corpus is GORILA through the LinearA Explorer: 1,720 documents and 1,007 distinct units, of which 156 have three or more attestations and at least one intact. That is the ceiling of the inventory: below three attestations there is no profile to measure.

The broken units. The words field of the edition preserves the breakage sign and the transliteration loses it. Recovered, **276 of the 783 syllabic units (35%) are broken in every attestation:** JMA-TE-RE is not MA-TE-RE. Three of the eight lexical coincidences with Linear B in a first comparison were fragments, and of Davis’s root/ending pairs, five of eight endings fall below three roots once only intact units count (-JA, -ME and -TI remain). The whole inventory is computed over intact units.

The gaps. The material culture of Haghia Triada includes copper ingots, honey and fish. No sign with a figure has the profile of those things (low whole numbers or weight for metal, a liquid fraction for honey), *118 is the weight unit and accompanies cyperus and sesame, and Younger’s lexicon does not mention them. **The tablet archive does not record them.** The contrast with Linear B is sharp: 1 of 1,480 logogram occurrences against 14.3%.

2.2 The instruments and their error rate

Twelve instruments, each with its null and its measured rate. Those that support the inventory:

instrument	what it measures	null	error rate at 0.05
quantity profile	fraction of a logogram’s figures against the base rate (27%), in the direction the class predicts; size of the figures	fractions at the base rate	4.6% (worst case, direction chosen a posteriori)
header	first position on a tablet against the base rate (11.8%)	binomial	positional: 3-5%
subtotal	figure smaller than that of the preceding	binomial	(within positional)

instrument	what it measures	null	error rate at 0.05
	commodity against the base rate (52%)		
qualifier	syllable added to two or more distinct commodity signs	rule, not test	—
stratified co-occurrence	two units in the same document, preserving site and margins	bipartite by site	5.5%
adjacency	two contiguous units, over eligible pairs	permutation	2.5%
name matching	exact coincidence with the Linear B lexicon by length	syllable bigrams of the corpus itself	(by length; 2 syllables = chance)
arithmetic	the figure after the unit equals the sum of those preceding	permutation	2.5%

An instrument is calibrated on the material it is applied to: the profile rate was measured by shuffling the fractions of already-read logograms at the base rate; the co-occurrence rate over words and logograms of the corpus itself (8.0% unstratified, 5.5% stratified by site: the site was the confounder).

2.3 The inventory

Sixty-five units: 25 established (field consensus confirmed by our test), 35 proposed (our test with no prior consensus, or prior art with no test until now), 5 disputed (a source against). Status, attestations, reading, prior source in favour, and number of items of evidence against:

unit	status	att.	reading	prior art	against
Arithmetic					
KU-RO	established	37	total	Godart & Olivier / field consensus, Sc	
KI-RO	established	16	deficit	Schoep, Uchitel	
DA-I	proposed	2	a total, of a kind distinct from KU-RO	—	

unit	status	att.	reading	prior art	against
PO-TO-KU-RO	established	2	grand total	Schoep	
Qualifiers and grades					
KU	established	170	as an added syllable, a qualifier of four commodities (GRA+KU, TELA+KU, *188+KU, C	—	
KA	proposed	169	as an added syllable, a qualifier of wine and of persons (VIN+KA, VIR+KA); alone,	—	
OLE+U	proposed	22	a type of oil counted in whole units (or a counted unit of oil: whole numbers, see	—	1
OLE+MI	proposed	19	a grade of oil; listed together with OLE+DI in every document where it carries a f	Younger	1
KI	established	19	as an added syllable, a	Del Freo, via Nosch and	

unit	status	att.	reading	prior art	against
			qualifier of oil and 316 (OLE+KI, 316+KI) and of wool (K	Weilhartner	
CYP+D	proposed	19	a fixed-quantity allocation of cyperus: a standard half-unit issued per person or	—	
PA	proposed	16	qualifier (a grade or type) shared by grain, cyperus and 304: GRA+PA, CYP+PA, 30	—	
OLE+DI	proposed	12	a grade of oil; listed together with OLE+MI in every document where it carries a f	Younger	1
TU	established	11	qualifier (a grade or type) restricted to the olive family: OLE+TU and OLIV+TU	Younger	
RA	proposed	9	as an added	—	

unit	status	att.	reading	prior art	against
			syllable, a qualifier of oil and wine (OLE+RA, VIN+RA)		
OLE+TA	proposed	8	a grade of oil recorded in small fractional amounts	Younger	
GRA+KU	proposed	7	GRA with the qualifier KU (a grade or type)	Acedo (proposer, extrapolation)	
VIN+RA	proposed	4	wine with the qualifier RA (a grade or type), parallel to OLE+RA	Acedo (proposer, extrapolation)	
MA-RU-ME	proposed	3	wool with the qualifier ME (a type or grade)	Del Freo, via Nosch and Weilhartner	
*188+KU	proposed	2	*188 with the qualifier KU (a grade or type)	Acedo (proposer, extrapolation)	
KI-MA-RU	proposed	1	wool with the qualifier KI	Del Freo, via Nosch and Weilhartner	
TELA+KU	proposed	1	TELA with the qualifier KU (a grade or type)	Acedo (proposer, extrapolation)	

**Headers and
transaction**

unit	status	att.	reading	prior art	against
TE	established	58	transaction term set between a party and the commodity on the tablets; on the nodu	Schoep, Younger	1
SA-RA ₂	disputed	20	transaction term, Haghia Triada only; possibly the ethnic of the site	Schoep, Uchitel	1
SA-RA2	proposed	20	the term that opens a commodity section: it stands before the goods and is not its	—	2
A-DU	established	10	heading term (record label) of HT tablets; heads the lists of HT 86 and HT 95 amon	Younger	
*307	proposed	6	occupies the header frame between the record label and the list, the same slot TE	—	3
KA-PA	established	6	heading term of HT tablets	Salgarella	

unit	status	att.	reading	prior art	against
			(HT only); KA-PA-QE is its extended form (Salgarella 20		
JE-DI	proposed	4	heading term (first position in 3 of 4 tablet attestations)	—	
A-KA-RU	established	3	heading term of HT tablets (HT 2.1, HT 86a.1, HT 86b.1 as second heading)	Younger	
DA-QE-RA	proposed	3	heading term of HT tablets (HT 6a.6 as a second heading, HT 120.1)	Younger	
KI-RI-TA ₂	proposed	2	heading term (first position in both attestations)	—	
Parties in the account					
SA-RU	proposed	6	a party in the account, not a variety of cereal: it recurs with five others	—	1

unit	status	att.	reading	prior art	against
			across		
DA-ME	proposed	4	a party in the account, not a variety of cereal: it recurs with five others across	—	1
MI-NU-TE	proposed	4	a party in the account, not a variety of cereal: it recurs with five others across	—	1
PA-TA-NE	proposed	3	a party in the account whose goods were sealed: it is a list entry on the tablets	—	1
QE-RA2-U	proposed	0	a party in the account, not a variety of cereal: it recurs with five others across	—	1
Commodities, livestock and persons					
NI	established	76	figs (the logogram NI; ni- as the first syllable of the Minoan	Neumann; Salgarella and Petrakis (eds.	

unit	status	att.	reading	prior art	against
			word)		
GRA	established	62	grain (cereal), the bulk dry commodity	Ventris and Chadwick	
VIN	established	53	wine, a liquid commodity	Ventris and Chadwick	
CYP	established	52	cyperus (*303), a fine commodity measured in small fractional amounts	Ventris and Chadwick	
OLIV	established	24	olives, the fruit as distinct from its oil	Ventris and Chadwick	
OLE	established	22	olive oil, a liquid commodity	Ventris and Chadwick	
*22F	established	14	goat, female; Linear B CAPf	GORILA / Ventris and Chadwick	
*308	proposed	12	fatty product derived from the olive	Koh & Birney, Younger	
*305	proposed	12	a category of persons	Younger	
*21M	established	8	sheep, male (ram); Linear B OVISm	GORILA / Ventris and Chadwick	
*23M	established	7	ox/bull; Linear B BOSm	GORILA / Ventris and Chadwick	

unit	status	att.	reading	prior art	against
*21F	established	4	sheep, female (ewe); Linear B OVISf	GORILA / Ventris and Chadwick	
*131B	proposed	4	a variant of the wine sign (AB 131b), a type of wine	GORILA	
*22M	established	3	goat, male; Linear B CAPm	GORILA / Ventris and Chadwick	
*309	proposed	2	counted by pieces and not in bulk: none of its ten quantities takes a fraction, wh	—	2
VIR	established	0	person (man), the unit counted in personnel lists	Ventris and Chadwick	
Other functions					
JA-SA-SA-RA-ME	proposed	7	does not designate the offering or the object	Finkelberg	
*306	disputed	7	possibly the sign for woman	Ferrara, Montecchi [3] & Valerio, Uchitel	2
KU-NI-SU	disputed	5	first name in a list; the Semitic	Younger, various, cited by Younger	1

unit	status	att.	reading	prior art	against
			reading as emmer is proposed and contested		
U-NA-KA- NA-SI	disputed	4	the term for what is offered, on the strength of a single tokenisation (SY Za 2) t	di Mino	3
Place names and personal names					
KI-DA-RO	proposed	2	a personal name shared with the Linear B tablets of Knossos (Godart 1984)	Godart	
PA-I-TO	established	2	Phaistos, the place name (Linear B pa- i-to)	Godart	
DA-I-PI-TA	proposed	2	a personal name shared with the Linear B tablets of Knossos (Godart 1984)	Godart	
I-TA-JA	proposed	1	a personal name shared with the	Godart	

unit	status	att.	reading	prior art	against
			Linear B tablets of Knossos (Godart 1984)		
SU-KI-RI- TE-I-JA	proposed	1	formed on the stem of Sybrita, or sharing its lexicon	Davis	
DA-MA-TE	disputed	1	coincides with Linear B da-ma-te; read as Demeter by some, contested	Davis, Duhoux, Owens	2
SU-KI-RI- TA	established	1	Sybrita, the place name (Linear B su- ki-ri-ta)	Godart	
I-JA-TE	proposed	1	coincides with Linear B i-ja-te (PY Eq 146); Younger: “of/from I- JA”	Godart, Younger	
SE-TO-I-JA	established	1	Setoia, the place name (Linear B se- to-i-ja; Archanes or Iouktas?)	Godart	

Four things are visible in the table as a whole.

Fourteen with no prior source. DA-I as a total of another kind; KI-RI-TA₂ and JE-DI as headers; VIN+RA parallel to OLE+RA; KU as a qualifier of four commodities and PA of three; KA of wine and of persons; CYP+D as cyperus in minimal amounts; OLE+TA as fine oil; OLE+DI and OLE+MI as paired grades; JA-SA-SA-RA-ME as what it is not. They are functions, and several are the same finding seen across several units.

Forty-two with prior art, where the measurement is what is new. VIR only in whole numbers (2 of 36 with a fraction, $p = 0.0008$); GRA in large whole numbers (12 of 90, maximum 976); CYP almost always in fractions and small amounts (39 of 64, maximum 23); TE in the divider-TE-divider-commodity pattern in 10 of 24 against 2.5% ($p = 10^{-10}$); livestock with a sex mark in whole numbers (2 of 28, $p = 0.007$); A-DU in first position in 7 of 10 ($p < 0.0001$). And in the other direction: six of Younger's "headers" with zero occurrences in first position, KU-NI-SU as emmer disputed, and "*304 always between OLE and OLIV" false as stated (6 of 25).

No words. All sixty-five are function: what the unit does in the account. For the sixteen signs of commodity, livestock and persons, the reading comes from pictographic identity with Linear B, and our test says the administration counted them as that class of thing is counted. No syllabic group has a translation: KU-RO is "total" because it sums, not because we know how "total" was said in Minoan.

What was tested and not fixed. The list names, the most numerous class, are judged only by exclusion: no fractions of their own, no added syllable, not a header, not a total; and that exclusion leaves a personal name and a commodity written in syllables looking identical. The nodule signs have their structure fixed, and what the reading of those signs awaits is the type of nodule per document [9], which is not in open source; and not their reading. The five units on the edge of two runs are not filed: a bare p of 0.01 to 0.04 with 50-80 tests is what chance produces.

2.4 How it was reached, and why the number stops

The inventory rose from 18 to 65 with the same procedure over the same units: the parallel with Linear B, the extrapolation of structures (qualifier, subtotal, header), name matching against a null, and two stages with a language model in the loop. Those two stages were measured: reading passages of the literature, the model invents non-existent units 22% of the time; proposing a reading from each unit's dossier, 1%. Its contribution was routing at scale, not the idea. A second pass over the remaining units found nothing.

That is the ceiling, and it should be stated precisely. It is not the ceiling of the problem: it is the ceiling of what can be **verified** with this corpus. The accounts have few functions and almost all are fixed; those that remain await an external datum; and the step from function to word requires a judge the corpus does not contain, because nothing in it can show that "KU-NI-SU means emmer" is false. That judge would be a bilingual or new running text, and those come out of the ground.

2.5 The opening position: a function that belongs to a place, not to a unit

The inventory above was built unit by unit. That pace has an arithmetic ceiling: only 148 units have two or more intact attestations and only 68 have three or more, so most of the corpus will never assemble enough evidence on its own. There are, however, functions that belong not to a unit but to **a position**, and fixing the position classifies at once every unit that occupies it.

Comparing, over the tablets, the first intact syllabic unit of each with all the others:

	units	with a figure following
the one that opens the tablet	184	53 (29%)
all others	464	334 (72%)

$p = 8 \cdot 10^{-34}$. The opening position takes no quantity: it is not an entry in the account, it is the label of the record.

One hundred and forty-one distinct units occupy that position; ninety-five never take a figure there, and **eighty-six of them had no function in the inventory**. The nine that did, fixed one by one above (A-DU on eight tablets, SA-RA₂ on six, KA-PA on four, A-KA-RU, JE-DI and KI-RO on three, and three more), all fall inside the class: the procedure recovers what was already established and adds the rest. The inventory thus rises from 65 readings to 151.

Two cautions are filed with the result. The remaining 29%, the fifty-three openings that do take a figure, are not labels: the rule belongs to the position and not to each case, and each of the eighty-six inherits the function with the force of the class and not with that of an individual reading. And knowing that a unit labels a record does not say whether it names a place, an official, a month or an operation.

The other positions do not yield. Tested against the same base rate: the second unit (58% with a figure, $p = 0.37$), the third (65%, $p = 0.88$), the one that closes the tablet (59%, $p = 0.47$) and the one following an already-read unit (56%, $p = 0.29$). None departs from it. And one test was discarded as circular: “the unit preceding a logogram” gives 0 of 100 with a figure and a p of $3 \cdot 10^{-40}$, which means nothing, because if the next token is a logogram then by construction it is not a figure. The opening is the only class-level function this corpus sustains, and it is so because every tablet instantiates it exactly once over 184 tablets.

2.7 The scope of the inventory: which genre each reading was fixed on

Part 4 states that the functions were fixed on Haghia Triada and weaken outside it. A second limit of the same kind has to be declared, and it came to light on reading Finkelberg [12] (2001).

The Linear A corpus holds at least three genres that do not behave alike: the accounting archive of tablets, the sealing system of nodules and roundels, and the votive inscriptions on stone vessels. Counted by the dominant genre of their attestations, the sixty-five readings fixed one by one divide as follows:

genre of the attestations	readings
accounting archive	51
votive, on stone vessels	7
sealing supports	4
other	2

Fourteen of the sixty-five therefore describe something other than the account, and among them are JA-SA-SA-RA-ME, U-NA-KA-NA-SI and DA-MA-TE, which belong to the libation formula, and TE, KU and KA, whose dominant use is on sealings.

Why it matters beyond bookkeeping. Finkelberg (2001) builds a fourteen-point morphological profile of Minoan and concludes that the language is Anatolian, with Lycian as its descendant. Eight of her fourteen features rest on the sequence A-TA-I-*301-WA-JA, whose eleven attestations are **all on stone vessels**. Her profile therefore describes the votive genre; this inventory describes the account. The two are not in competition over most of their extent, and saying so is not a concession but a statement of what each measures.

They do meet at one point. She reads *a-du* as the third person imperative of an Anatolian verb *a-* “to make”; this inventory has A-DU as a heading term on ten tablets. Measured, her paradigm — *ja-di* (1 attestation), *je-di* (4), *a-da* (1), *a-du* (10), *a-ra-u-da* (1), with *a-di* **unattested** — occupies the first syllabic position of its document in **12 of 17 attestations against a base rate of 0.405, $p = 0.012$** . A verb does not systematically head its document; a record label does, and §2.6 measures that class at $p = 8 \cdot 10^{-34}$. On this point the distributional reading holds, and it holds inside the archive, which is where both attestations of the disagreement lie.

2.8 What the combination of picture and syllable says

The corpus has two classes of sign: the logogram that draws the commodity and the syllabogram that writes sound. Ligatures bind them — GRA+PA, OLE+KI, CYP+D — and this inventory has treated the two classes separately, measuring some as commodities and others as qualifiers. The relation between the classes had not been measured.

Thirty-nine distinct ligatures bind 26 syllabograms to 18 logograms. The question is whether the bound syllabogram belongs to its commodity or circulates among several: if it abbreviates something about that commodity it should be specific, and if it is a general grade mark it should circulate. The null reassigns the syllabograms to the logograms preserving how many ligatures each commodity carries, 2,000 repetitions.

	observed	null
syllabograms bound to one commodity only	16 of 26 (62%)	25%
bound to two or more	10	19.4 on average, minimum 15

p < 0.0005. The bound syllabogram goes with its own commodity far more than chance allows, which is what the acrophonic reading predicts: Linear B writes CYP+KU for *ku-pa-ro*, and the editors of the Knossos ivory inscriptions describe the same device for their vessels [6].

And the ten that do circulate are the ones this inventory had fixed. They are D, E, K, KA, KI, KU, MI, PA, QE and RA, and among them are precisely the entries filed above as general qualifiers: KU across grain, *188 and textile; PA across *304, cyperus and grain; KI across *316 and oil; KA across wine and persons; RA across oil and wine. Those entries are measured one by one and stand, but their standing changes: **they are the exceptions of a system whose rule is the opposite**, and the inventory had recorded the minority without seeing the majority.

One reading error is worth recording with the result. On first seeing the list of ten shared syllabograms the immediate conclusion was that the system was one of general grades and that acrophony did not hold. It was the reverse, and the null said so: ten shared looks like many until one sees that chance would give nineteen. **A count without its null does not merely exaggerate an effect; it can invert its sign.**

What this does not give is what any syllabogram abbreviates. That OLE+KI is specific to oil does not say what KI is, and no phonetic value is fixed by it. What is established is a property of the script: **Minoan writes the type of a commodity by binding to its picture a syllable that belongs to that product**, and that is a rule of the writing system, measured, that requires reading nothing.

What the ligature is worth as a constraint on values, priced. If the bound syllabogram abbreviates something about its commodity, then its phonetic value is constrained: it must be the first syllable of a word in that commodity's field. That is a constraint on a sign's value obtained without comparing whole words, and no decipherment method uses it, because none looks at the logograms.

Nineteen syllabograms bind to a single commodity whose semantic field can be defined. Against the glossed lexicons of Akkadian, Ugaritic and Sumerian, each admits between 15 and 33 possible values under the acrophonic hypothesis, against the 74 a free sign admits. Under that hypothesis the constraint would be worth 36 bits of the 587 that a decipherment costs. **It was tested on Linear B before being claimed, and it fails there.** Where every value is known, the bound syllabogram is the first syllable of a word in the commodity's field in 2 cases of 10 (null 0.9, $p = 0.22$), and of a word adjacent to the logogram in 4 of 11 (null 3.23, $p = 0.41$). OLE+PA is *pa-ko-we* "sage-scented", OLE+A is *a-ro-pa* "unguent", CYP+KU is *ku-pa-ro*: each has its explanation and none follows a rule a model could apply without already knowing the answer. The distributional fact stands; the constraint on values does not, and the 36 bits are withdrawn.

Part 2 bis. The formats of the tablet

Motif discovery, which in bioinformatics locates conserved patterns with variable positions across few sequences without knowing what they mean, is applied here to administrative

formats. Each tablet becomes a sequence of functional categories (W syllabic word, C commodity, N figure, f fraction, T total, L single sign, | divider) and recurrent n-grams are counted against a first-order Markov null preserving category frequencies and the transitions between them. 308 tablets, 80 runs of the null.

The positive control is supplied by the procedure itself: the motif W | L | C N occurs seven times against 0.48 expected, and it is the transaction template that §1 fixes by reading (SA-RO · TE · VIN, on HT 6a, 14, 17, 19, 21). The method recovers it unprompted, and extends it at both ends: the full template is party, divider, transaction term, divider, commodity, quantity.

Four further formats, none previously described as a genre:

motif	occurrences	null	documents	what it is
W f W f W f	6	0.01	HT 8a, HT 98a, PE 2	list of parties, each with a fractional quantity and no commodity named
C f C f C f	14	0.14	HT 23a, 35, 50a, 91, KH 8, KH 15	list of commodities, each in fractions
W N f W N f	5	0.45	HT 9a, 19, 36, 104	parties with mixed quantities
W W C N	5	0.19	HT 28b, 96a, KH 5, KH 7a	two separated words before the commodity

The format C f C f C f is Khania's, appearing in two of its six documents, and anticipates by an independent route what §4 establishes about that archive. A classification error corrected during the test should be declared: the edition writes some values as approximations (“ $\approx \frac{1}{6}$ ”) and the classifier did not recognise them as fractions, which generated a spurious motif.

Part 2 ter. The list entries: groups, modules and metrology

The most numerous class of the corpus, the entries followed by a figure, was also the only one without an instrument. Four tests failed: the profile of endings within the class (one ending shared by two roots, null 0.4, $p = 0.36$), concentration by site (an artefact: 87% in a single site against 32% expected, but the list is headed by KU-RO and KI-RO, and within Haghia Triada none of the twenty entries concentrates on a scribe, $p = 1.00$), the stability of quantities (entries vary **less** than commodities, 0.95 against 1.54, the opposite of what was predicted) and the suffix

-JU (11 of 317 against 1 of 344, $p = 0.0004$, which passes correction but with no bare root to confirm it). The fifth works.

2 ter.1 The entries recur in groups, and this was seen in 1974

Prior art, checked item by item. Packard [11] (1974, pp. 62-63) first observed that HT 86 and HT 95 form a connected pair. Uchitel and Finkelberg [10] (1995) set out three tables of recurrent items by inspection. Compared against the groups measured here:

- **Their Table I is group C exactly**, down to the six items *ku-ni-su*, *sa-ru*, *di-de-ru*, *qa-ra₂-wa/qe-ra₂-u*, *da-me* and *mi-nu-te* under the headings *a-ka-ru*, *a-du* and *da-du-ma-ta*. That group is theirs, and the contribution here is only its measurement.
- **Their Tables IIa and IIb are not groups A and B.** Their IIa holds fifteen items and shares four with group A; their IIb holds ten and shares three with group B — and four more with group A. The two partitions overlap substantially and do not coincide: theirs is organised by document, ours by co-occurrence above a null, and ours separates A from B where theirs distributes members of A across both tables.

Two of their observations were verified directly against the corpus and hold: *ku-ni-su*, an item of group C, does head HT 10a; and *ki-ki-ra-ja*, which they read as a form related to *ki-ro*, does head HT 85b.

Their reading of the items as men from six villages under different headings is stated by inspection, without a null, and is recorded here as prior art rather than as support. The agreement with the conclusion reached below is worth noting and is not evidence: two routes can share a premise about what a list of six with quantities must be.

If the entries are parties of an administration, the same ones should reappear together from one document to another. Over the 56 documents with three or more entries, setting aside the units already read, the pairs coinciding in two or more documents number **35 against 8.2 expected** under a null that reassigns entries preserving the size of each document and the frequency of each entry (600 repetitions, maximum of the null 18). $p < 0.002$. The inspection of 1995 is confirmed at a measured effect size.

Four groups, each with its profile:

group	members	documents	what it records
A	DA-RE, DA-RI-DA, DA-SI-*118, KU- ZU-NI, TA-NA-TI, TE-KI, U-*325-ZA	HT 7a, 10a, 10b, 13, 85a, 122a	personnel, quantities from 1 to 72
B	*306-TU, *324-DI- RA, DI-NA-U, PA- DE, QE-PU, TA-I-	HT 9a, 9b, 122a	wine and aromatics, 1 to 4

group	members	documents	what it records
	AROM		
C	DA-ME, DI-DE-RU, MI-NU-TE, QE-RA ₂ - U, SA-RU	HT 86a, 95a, 95b	grain, 10 and 20
D	*28B-NU-MA-RE, QE-SI-ZU-E, SI-PI- KI	ZA 4a, 5b, 15a	Zakros

They are sets, not sequences: comparing across documents whether two members keep the same relative order, 32 agreements of 55, 58%, against the 50% of free order ($p = 0.14$). Group D, entirely from Zakros, rules out the phenomenon being an artefact of Haghia Triada.

Group C receives fixed quantities. Its mean coefficient of variation is 0.34 against 0.75 for the 58 entries of the corpus with two or more quantities ($p = 0.021$ over 5,000 samples of five entries). And the same set of six reappears under three distinct headings: DA-DU-MA-TA on HT 95a, A-DU on HT 95b, A-KA-RU and A-DU in the two sections of HT 86a.

Whether they are parties receiving or varieties of cereal was open until the attestations were reviewed one by one. The aggregate test does not decide: a variety does not receive olives or cyperus while a party does, and two of six have two or more associated commodities against a base rate of 0.116, **$p = 0.147$** . What decides is four individual attestations incompatible with the alternative. SA-RU takes the figure 1 in the deficit list of HT 94b beside four units that behave as parties, one of which (PA-TA-NE) is independently attested on a sealed nodule, and it takes OLIV 16 on HT 123+124a. MI-NU-TE heads a record of cyperus on HT 106. DA-ME is followed on HT 120 by two distinct grain ligatures, GRA+K+L 74 and GRA+PA, which a grain variety cannot be. QE-RA₂-U stands immediately before KI-RO 197, the largest deficit of the archive. **They are parties.**

Uchitel and Finkelberg (1995) read the same items as men from villages, by inspection. One datum of theirs is checkable and checks out: *ku-ni-su* heads HT 10a, which a cereal name would not do. KU-NI-SU remains filed as disputed, since its own attestations are otherwise all in grain contexts.

2 ter.2 Allocation is modular

Over the 119 tablets with four or more whole figures, with a null that shuffles all the figures of the corpus between documents preserving how many each carries (500 repetitions):

- **Eight have every figure a multiple of the smallest, against 1.9 expected** (maximum of the null 6, $p < 0.002$). The modules: 30 on HT 146; **10 on HT 95b and HT 86a**, which are group C's; 5 on HT 31 and HT 47a; 2 on HT 39, HT 41a and PH 28b.
- **Fifty-one repeat one figure three or more times, against 36.0** (maximum 46, $p < 0.002$).

That the two tablets with module ten are group C's is the independent confirmation that result lacked. The archive did not assign free quantities: it allocated in units.

2 ter.3 Each commodity has its own metrology

Over the 645 whole figures associated with a commodity, with a null that mixes them between commodities preserving how many each has:

commodity	figures	median	multiples of 10	null	p
GRA	151	10	34%	17%	<0.002
VIN	68	5	18%	17%	0.50
VIR	118	10	17%	17%	0.61
NI	77	3	14%	18%	0.85
OLE	120	3	8%	17%	0.998
OLIV	25	3	4%	18%	0.994
CYP	49	3	4%	17%	1.000

Grain is counted in tens and fine products are not. There are at least two regimes of measure: bulk in tens, and oil, olives and cyperus in small units and fractions. That gives the cause of what §4 describes: Khania never sums and records in fractions because its dominant commodity is cyperus, not because it counts differently. What this measurement does not fix is what a unit is worth; Corazza and colleagues [5] (2021) fixed the values of the fractions, and those of the whole numbers remain unfixed.

Part 3. The aromatics file

3.1 The identification of *303, which is disputed and is measured here

The reading of *303 as cyperus, which this work follows after the edition consulted, is not unanimous. Goold and Pope (1955), followed by Uchitel [4] (2003), identify it with barley; Palmer (1995) proposed emmer wheat. All three identifications rest on the shape of the sign, on its relative frequency, or on comparison of resources between archives, and none on its quantitative distribution within the corpus. It is worth measuring, because the premise conditions everything that follows.

sign	n	median	maximum	records with a fraction
GRA (grain, identified by inheritance from	90	15.62	976	13%

sign	n	median	maximum	records with a fraction
Linear B [13])				
olive	21	10.00	93.5	29%
wine	34	3.62	104	29%
oil	100	3.62	100	26%
figs	50	2.00	75	34%
*303	64	0.62	23	61%

Most records of *303 are of less than one unit and six in ten carry a fraction, against the grain of the same archive, whose median is twenty-five times larger and which records fractions in one case in eight. Company differs likewise: *303 does not coincide with the olive in any of its 53 documents, while grain does so in twelve of seventy-two.

The control is in Linear B, where the identification is not disputed: there grain reaches 10,300 units and cyperus ten. The profile of *303 (median 0.62, maximum 23, mostly fractions) is the second and not the first.

This does not prove *303 is cyperus. It proves it **is not counted as a cereal**, which rules out the most cited alternative reading and leaves the field to substances dosed in fractions. It also answers the argument from frequency: *303 can be the most recorded commodity of Khania without being a grain, if Khania was a centre of aromatics production, which is what the chemistry of the region's workshops indicates.

One datum tells against this reading and must be recorded. On the four tablets that combine a personnel section with a commodity section (HT 27a, HT 89, HT 94a, HT 100), the commodity section always follows the same order, *303 followed by figs and wine, with additions at the end; it is the sequence Uchitel (2003) calls Standard II. That *303 systematically heads that sequence places it in the position of the principal product, which is the position a cereal would occupy in a ration list. Position alone does not decide, but it is an argument for the alternative reading and not for the quantitative profile set out above.

3.2 Oil divides into two classes

The oil logograms of Linear A divide into two groups by their quantities and their company. OLE+U, plain OLE, OLE+KI, OLE+MI, OLE+DI and OLE+TU are recorded in whole units (medians between 3 and 7, up to 54) and share documents with cereal, olives, figs and *304, above chance in every case: these are the oils of estate production. OLE+TA, OLE+RI and OLE+NE are recorded in fractions and share documents with *303 (OLE+TA in 4 of 7 documents, $p < 0.001$), with *308, MI+JA+RU and the ligatures QA₂+[]+PU and QA₂+[]+RE, and almost never with cereal or olives. The second group corresponds in function to the qualified oils of the Fr series at Pylos (*ku-pa-ro-we*, *wo-do-we*, *pa-ko-we*), with the qualifier written as a

bound sign instead of as a word — a device Linear B also uses in CYP+KU and PYC+KU, and which the editors of the new ivory inscriptions from the Knossos cult centre describe in the same terms (Kanta, Nakassis, Palaima and Perna [6] 2024). The values of TA, RI and NE are unknown, and OLE+TA, the one most often recorded with *303, is the natural candidate for cyperus oil.

3.3 The ingredient block

The four signs accompanying the qualified oils form a closed block. Over the 1,719 documents of the corpus, *308 (8 documents), MI+JA+RU (7), QA₂+[*]+PU (5) and QA₂+[*]+RE (5) co-occur with *303 and with OLE+TA, OLE+RI and OLE+NE far above expectation (hypergeometric, $p < 0.001$ in every case), are always recorded in fractions, and *308 and MI+JA+RU never appear with cereal or figs. A fifth sign the first pass had included, *188, does not belong to the block: it is recorded in whole units, heads documents and appears with the KI-RO deficits, so it is another commodity and is withdrawn from the file.

A limitation of this test must be declared. The documents containing the block are richer in logograms than the corpus average (4.0 against 1.4), and that richness inflates any co-occurrence: the hypergeometric test controls for how many documents each sign appears in, but not for how many logograms each document carries (capture-artefact control, $r = 0.415$). What sustains the result is therefore not the raw co-occurrence but two things richness does not explain: that the fixed-margin null, which does preserve document sums, produces the same clustering; and that the contrast between what the block accompanies (oil and olive, 7 documents against 1.9 expected, $p = 0.002$) and what it never accompanies (textiles, zero of fourteen) is asymmetric, where richness would affect both alike.

The grouping of these documents is not new either. Uchitel (2003) classifies the Haghia Triada tablets into standard commodity sequences, and his Standard Sequence III gathers HT 23a, HT 32, HT 33 and HT 35, four of the five central tablets of this file, on a different criterion. And in the same work he identifies SA-SA-ME as the Minoan equivalent of Mycenaean *sa-sa-ma*, “sesame”; that identification is taken from him and not offered as our own. What this work adds to the grouping is its measurement.

3.4 The format: the Minoan spice lists

HT 16 deserves separate mention. Its entries are single signs used as commodity abbreviations followed by small fractions (WA $\frac{1}{3}$, *188+KU $\frac{3}{8}$, SA-PO ZE $\frac{1}{16}$), a format found among Linear A tablets only in HT 20 as well, written by the same hand according to Godart’s attribution (GORILA V [1], scribe HT 10), on which the scope of palaeography as a criterion is discussed by Salgarella and Judson [8], so that the format is in part a scribal habit. It is the format of the Ge series at Mycenae, in which condiments are recorded by acrophonic abbreviation with the smallest measures of the system.

3.5 Sesame in this file

SA-SA-ME (AB 31-31-13) is attested once, on HT 23a, with the fraction $\frac{1}{16}$, in the same type of document that supports the three previous cases. In the Ge series at Mycenae sesame, *sa-sa-ma*, occupies that same position, that of the smallest measures of the system, beside fennel, safflower and mint. Greek σήσαμον is a recognised Semitic loan (Akkadian *šamaššammū*, Ugaritic *ššmn*), and the Minoan form preserves the reduplication of the source with its own final vowel, which places it as a link in the Minoan channel that Melena and Piquero [7] (2015) postulate for commodity terms.

The case belongs to a different class from the previous three and is declared as such: it is not a name formed on a plant name but the word for the commodity itself. Its force is smaller: a single attestation, no family, no affixes and no possible co-occurrence null; and the ambiguity between commodity and party, which the seals decided in the case of KU-PA, cannot be decided here. What inclines towards the commodity is the format of the document, the absence of derivatives and of any appearance in a header, roundel or personnel list, and the exact parallel of the Ge series.

3.6 Discussion of the aromatics file

What is described is an administration of aromatics and perfumed oils: a class of oil measured in fractions and allocated together with cyperus and four fixed ingredients, in documents with their own format, in which the allocation header SA-RA₂ appears twice followed by cyperus and OLE+TA. The Mycenaeans, two centuries later, write the same thing with words: the oil is called “of cyperus”, “of rose”, “of sage”. The Minoans wrote it with ligatures, and that is why the file had not been seen: it is not in the vocabulary but in the signs bound to the logograms.

What cannot be decided. First, the values of TA, RI and NE: they are acrophonic qualifiers of something, and the corpus does not say of what; OLE+TA is a candidate for “cyperus oil”, and that is a hypothesis, not a reading. Second, the identity of *308, MI+JA+RU and the two QA₂ ligatures: their distribution places them in the file and nothing more. Third, whether the proportions are a recipe: the ingredients appear with equal quantities within the same document more often than the corpus quantities predict (0.26 against 0.06; $p = 0.0005$), but not more than a null restricted to fractions predicts (0.21; $p = 0.24$), so the question is undecidable with the legible quantities.

The Minoan channel, measured over the whole semantic field. The hypothesis that certain commodity terms reached Greek from the East through Minoan has been argued word by word. With the list of twenty Bronze Age perfume substances, closed by the Assur recipes and the Mycenaean series, it can be measured as a whole: 90% of their Greek names are not Indo-European (18 of 20), against 35% of a control set of twenty concrete nouns from other domains (7 of 20; Fisher exact test, $p = 0.0008$). And within the non-Indo-European ones, the pre-Greek outnumber the Semitic two to one. When the Greeks inherited the perfume industry they inherited its lexicon too, and that lexicon was neither theirs nor, for the most part, directly Eastern. A qualification is due: “pre-Greek” is not “Minoan”, since it designates the Aegean

substrate in general; what points to Minoan specifically is that six of those substances have an attested counterpart in Linear A itself.

What would refute this work. Three observations would. First, that residue analysis of processing vessels from a Minoan aromatics workshop finds no *Cyperus* markers (cyperene, α -cyperone, cyperotundone, rotundene) while finding those of the other proposed ingredients. Second, that a new tablet records *303 in large whole quantities beside olives or livestock, which would place it in the profile of a cereal. Third, that the equality of fraction between secondary entries turns out to be an artefact of fraction notation rather than a property of the documents.

Part 4. The archive of Khania: another format, another practice

Haghia Triada is 64% of the corpus, and everything above describes its archive. Khania, with 227 documents and 105 tablets, is the second usable series, and measured against its own base rates it turns out to be something else.

Khania never sums. None of its 105 tablets carries KU-RO, KI-RO, PO-TO-KU-RO or DA-I; at Haghia Triada 41 of 205 do (20%). At the HT rate, the probability of having none in 105 tablets is $7 \cdot 10^{-11}$. Across the whole Khania corpus, one document of 227 carries a summation term (0.4%, against 3.7% at HT). The difference is one of administrative practice, not of sample size.

Khania records something else, and in another way. Its dominant commodity is CYP (65 occurrences against 14 at HT), and 68% of its CYP records carry a fraction, against 36% at HT. Its tablets have 7.4 units on average against 13.8, a quarter of the syllabic groups per tablet (0.8 against 3.1), and smaller figures (median 3 against 5). It is an archive of fine distribution of fine products, against the bulk store of Haghia Triada. **§2 ter.3 gives the cause:** cyperus is not counted in tens anywhere in the corpus, so an archive dominated by cyperus records in fractions and has less to total.

What Khania does not have: headers of its own (with a base rate of first position of 15% over its 99 tablets of three or more units, only A-DU passes bare, and it is the header already fixed at HT) nor qualifiers of its own.

What Khania does give: CYP+D as a fixed-quantity allocation. At Khania, CYP+D carries the same quantity ($\frac{1}{2}$) in 12 of its 16 attestations with a figure, against a base rate of 22% for that figure among all quantified logograms of the archive (binomial $p = 10^{-5}$). No commodity of the Haghia Triada archive has a dominant quantity at that level. And it is issued after personnel entries: KH 7a gives VIR+*313b 10 followed by CYP+D $\frac{1}{2}$; VIR+*313b 4 followed by CYP+D $\frac{1}{2}$. It is one of the two quantities in the corpus that behave like a ration — the other being group C's grain in tens (§2 ter.1) — and it would be refuted by new documents giving CYP+D large or varied quantities, or with no persons around.

Consequence for everything above. The functions fixed in Parts 1 to 3 were fixed on the archive of Haghia Triada. KU-RO as a total is a reading of that archive; at Khania there is

nothing to check it against, and its absence there is not accidental. The inventory should state, entry by entry, in which archive each function was fixed, and general claims about “the Minoan receipt” hold for Hagia Triada and weaken outside it.

Part 5. Discussion

What this inventory changes is not the number of readings but their status. Each one carries a statement of what observation would refute it, and the dictionary refuses to file an entry without that line. Each instrument carries its error rate, and a rate that came out wrong is kept as a record of why the null was changed and not the instrument. Each claim with prior art carries the citation and the measurement together, and those that fall are filed as fallen, with the source against.

And what the inventory says about Linear A, read whole, is something a century of proposals had not said in this form: **we know what was counted and how the account was organised, we know three places, and we know no word.** The distance between an archive that is understood and a language that is read is exactly that, and no internal method closes it.

To that, the parts together add a picture none gives alone. The Minoan receipt is that of its period and region, with no feature separating it from Ebla or Pylos in what it does, which means the problem of Linear A is not in the administration, which is understood, but in the language, which is not. The inventory says how far understanding the administration goes: 151 functions, almost all an accounting archive has. The aromatics file shows that within that administration one can descend to the class of a product without reading a word. The groups, the modules and the metrology show that the administration allocated in quotas and that the quota depended on the commodity. And what remains open is what it always was, said more precisely: the step from function to word, for which this archive contains no judge.

Data and reproducibility

Corpus, code, null models, error rates and the manifest of every figure at github.com/BiomeMakers/kuro.

Statement of artificial intelligence assistance

The measurements, the null models and the drafting of this paper were carried out with the assistance of a language model (Claude, Anthropic), under the design, verification and responsibility of the author. Two stages of the procedure place the model in the loop, and its invention rate at those stages is measured and reported in §2.4.

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